

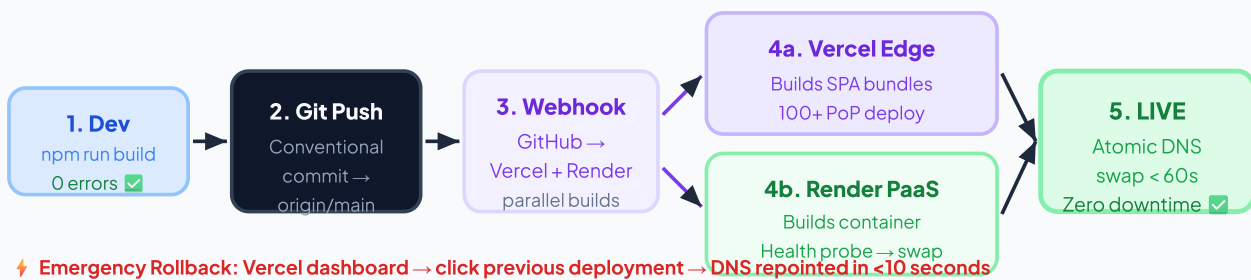
DevOps, CI/CD & Cloud Deployment

Zero-downtime continuous deployment: Git trunk-based development, Vercel Anycast Edge CDN, Render managed containers with rolling deploys, instant rollbacks <10s, and secret vault hygiene.

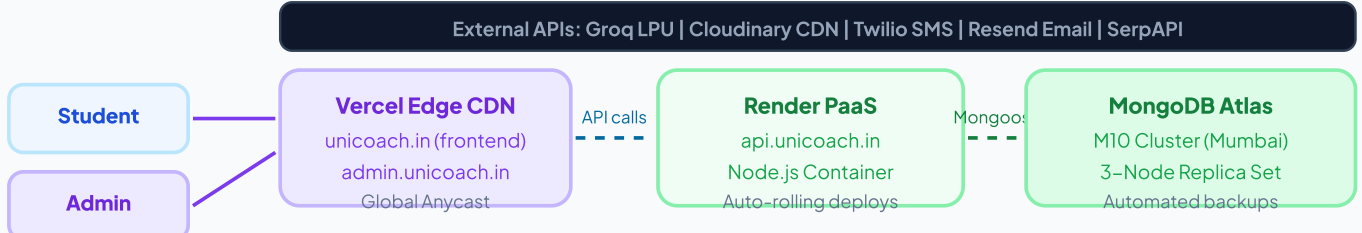
🌐 Frontend: Vercel Edge Network 🚀 Backend: Render PaaS ⏪ Rollback: <10 seconds

1. End-to-End Deployment Pipeline

🚀 ZERO-DOWNTIME CI/CD PIPELINE — LOCAL COMMIT TO GLOBAL PRODUCTION



🌐 PRODUCTION INFRASTRUCTURE MAP — UNICOACH CLOUD ARCHITECTURE



2. Technology Decision Framework

1. Vercel Anycast Edge vs Self-Hosted Nginx on EC2

FRONTEND HOSTING

WHAT IS IT?

A serverless static hosting network distributing compiled SPA assets across 100+ global Edge PoPs with automated SSL, compression, and immutable caching.

HOW DOES IT WORK & PROBLEM SOLVED?

An overseas student in London is routed to the nearest European PoP via Anycast DNS. TLS handshake and asset delivery occur within 30ms instead of traveling to a single origin server in India. Brotli compression reduces asset sizes by an additional 15–20%.

WHAT ALTERNATIVES EXISTED?

Self-hosted Nginx on EC2: Manual server setup, Certbot SSL renewal, DDoS firewall config, manual CDN invalidation, 40h/month ops overhead.

WHY DID WE CHOOSE THIS APPROACH?

Zero server management: push to Git → live globally in 60 seconds. Preview deployments on every PR. Instant atomic rollback. \$0 for startup traffic volumes.

2. Render Managed PaaS vs Kubernetes (EKS)

BACKEND COMPUTE

WHAT IS IT?

A managed container hosting platform that runs the Node.js API with zero-downtime rolling deployments, health check probing, and automatic container restart on crash.

HOW DOES IT WORK & PROBLEM SOLVED?

On new deployment: Render spins up the new container, polls `GET /api/health` until 200 OK, then atomically shifts the load balancer to the new instance. Old container only retires after all in-flight requests complete.

WHAT ALTERNATIVES EXISTED?

Kubernetes on EKS: Requires 3 control plane nodes, Helm charts, ingress controllers, and dedicated DevOps engineer. Overkill for <10K daily users.

WHY DID WE CHOOSE THIS APPROACH?

Right-sized reliability: container isolation, auto-scaling, and zero-downtime deploys without the operational overhead of self-managed Kubernetes clusters.

💡 Senior Interview Defense: "How do you handle deployments and infrastructure?"

"UniCoach uses a decoupled deployment architecture. The two React SPAs deploy to Vercel's global Anycast Edge Network — push to main, and assets are live on 100+ PoPs globally in under 60 seconds. The Node.js API deploys to Render's managed container PaaS with rolling deployments: new container spins up, health probe returns 200, load balancer atomically shifts. If anything breaks in production, I can rollback to any previous Vercel immutable deployment in under 10 seconds — no git reverting, no rebuild, just DNS pointer swap."